

ZIJIAN FENG

Singapore PR

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EDUCATION

- B.Sc. Honours in Computing Science, The University of Glasgow, 2016

RESEARCH INTERESTS

- **LLM/VLM Post-Training, Application and Evaluation:** synthetic data, agentic RL, evaluation of both content understanding LLM/VLM and agents
- **Computer Vision and Robotics:** 3D computer vision (scene understanding, novel view synthesis, depth estimation), large-scale image retrieval, object recognition (detection and segmentation), multi-task learning, deep model compression and quantization
- **ML Infra and Tools:** PyTorch, Triton

PROFESSIONAL EXPERIENCE

TikTok PTE. LTD.

Singapore

Staff Applied Scientist - Business Integrity

11.2023 - present

I lead a team of 7 full-time scientists and engineers to build the AI foundation for TikTok Business Integrity, with a focus on large-scale foundation model data, moderation intelligence, and agentic system evaluation. My team built fine-grained integrity knowledge and online synthetic data generation infrastructure at scale, upgraded moderation algorithms, and drove the agentic transformation of platform risk governance.

- **Agentic Ads Moderation System.** Built an agentic content moderation system for from scratch, worked end-to-end on problem definition, data curation, agentic SFT, agentic RL and serving. The system achieves significant gains in recognition performance through adaptive feature selection with tool calls, allows agile adaption to changes in moderation policies, while being cost-efficient with throughput >2 QPS per NVIDIA H20.
- **Large-Scale Synthetic Data for Foundation Model.** Constructed a >300-entity fine-grained integrity knowledge graph and an online multimodal synthetic-data pipeline, using multimodal embedding-based entity grounding to map real-world videos to granular risk concepts and automatically generate aligned QA, captions, spatial grounding, and agentic moderation tasks, enabling continuous high-quality data generation for foundation model training and evaluation. Our pipeline generates >2B tokens of data daily and leads to continuous improvement on our in-house foundation model.
- **Agentic System Evaluation.** Established large-scale evaluation infra, constructed benchmark datasets for internal risk-operation agentic systems. Our infra supports automatic multi-round conversation with user simulator, rollouts of trajectories in sandboxed environment with >20 internal skills, as well as comprehensive evaluation of agentic capabilities such as long-horizon tasks and memories.

Gaussian Robotics

Singapore

Staff Research Engineer

11.2021 - 11.2023

I built the technical foundation from scratch for machine learning model training, evaluation and deployment on low-compute robotics systems.

- **Large-Scale Multi-Task Learning.** Developed algorithms to run a dozen of deep-learning-based recognition tasks on edge devices (2 TOPS) and established MLOps for fast iterations of the model.
- **Model Development Toolkit.** Built infrastructure for model development, including self-supervised pre-training, supervised training, INT8 quantization (both PTQ and QAT) and runtime conversion.
- **End-to-End Testing System.** Built an end-to-end Hardware-In-the-Loop (HIL) testing framework with a web interface for pre-release checks for complicated software systems.
- **Foundation Models Research.** Developed a semi-automatic data annotation system based on the Segment Anything Model (SAM) and developed different fine-tuning techniques to improve the model performance on our dataset.

TikTok PTE. LTD.**Singapore**

Senior Research Scientist - AI Lab

11.2020 - 11.2021

Conducted cutting-edge research on novel view synthesis and worked on large-scale visual search:

- **Novel View Synthesis (Research Project).** Co-developed MINE - a novel method that performs dense 3D reconstruction from a single image with neural radiance field (NeRF) and multiplane images (MPI), our work was accepted to ICCV 2021 (<https://vincentfung13.github.io/projects/mine>).
- **Visual Search for TikTok E-Commerce.** Developed the visual search embedding model for large-scale product retrieval from TikTok videos. Co-designed the search pipeline and re-ranking algorithms.

ViSenze PTE. LTD.**Singapore**

Senior Research Engineer

06.2019 - 06.2020

Worked on large-scale visual search for e-commerce domain:

- **ICCV 2019 DeepFashion2 Clothes Retrieval Competition.** Developed the retrieval model and the reranking pipeline, which helped us secure the 1st place of the competition.
- **Image Retrieval Algorithms.** Trained most of the company's online image retrieval models, and developed a training codebase for the whole algorithm team. Replaced all online models with efficient architectures (MobileNet, ShuffleNet) without any performance drop. Developed a multi-task embedding training method to train models with high category relevance while maintaining the same-product image retrieval performance.
- **Large-Scale Distributed Training.** Developed a model training solution with a mixture of data parallelism and model parallelism to train discriminative embedding with arc softmax losses with millions of classes.

Tuputech PTE. LTD.**Guangzhou, China**

Lead Research Engineer

06.2016 - 06.2019

Led a team of 5 research engineers for large-scale pedestrian recognition (Re-ID) and tracking:

- **Person Re-Identification and Tracking.** Led a team and built a system for large-scale pedestrian recognition and tracking across multiple cameras. Established data-driven workflow for person re-identification model development and designed the tracking pipeline.
- **Image Classification.** Developed and maintained a backend model training platform for classification-based models with TensorFlow for training / evaluating image classification models, which allowed the solution engineers in the company to quickly train / evaluate a model with a few clicks.

PUBLICATIONS

- Yiheng Lin, Siyu Jiao, Xiaohan Lan, Wei Zhou, Qi She, Fei Yu, Heyun Chen, Zhengwei Wang, Jinghuan Chen, Moran Li, Yingchen Yu, **Zijian Feng**, Yao Zhao, Yunchao Wei, Yujie Zhong. TextSculptor: Training and Benchmarking Scene Text Editing. Preprint 2605.21090.
- Jiaxin Li*, **Zijian Feng***, Qi She, Henghui Ding, Changhu Wang, Gim Hee Lee. MINE: Towards Continuous Depth MPI with NeRF for Novel View Synthesis. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), October 2021. (*Equal contributions)
- Yike Yun*, Linjie Hou*, **Zijian Feng***, Wei Jin, Yang Liu, Heng Wang, Ruonan He, Weitao Guo, Bo Han, Baoxing Qin, Jiaxin Li. A Deep-Learning-based System for Indoor Active Cleaning. In Proceedings of the International Conference on Intelligent Robots and Systems (IROS), October 2022. (*Equal contributions)

COMPETITIONS AND ACADEMIC SERVICES

- Competition: 1st place in ICCV 2019 DeepFashion2 clothes retrieval challenge
- Conference reviewer: ECCV'22, ICRA'22, CVPR'23, ICCV'23
- Journal reviewer: IJCV